

Medvale Internal Medicine

Chapter 12C: Preventive, Geriatric, and Palliative Ambulatory Medicine

Health maintenance, screening, vaccination, geriatric syndromes, deprescribing, goals of care, and outpatient safety algorithms

Chapter purpose

This capstone ambulatory chapter shifts from disease-by-disease management to longitudinal care. It covers preventive medicine, risk reduction, adult vaccination, cancer screening, geriatric assessment, falls, delirium, dementia, depression, polypharmacy, incontinence, pressure injury prevention, nutrition, osteoporosis, palliative care, hospice, informed consent, symptom control, and integrated outpatient algorithms. It is written for high-yield clinical reasoning and Graph-RAG extraction: each section links a patient state, a risk factor, a screening or assessment tool, a management threshold, and red flags that require escalation.

The uploaded Step-Up ambulatory material frames ambulatory medicine around common outpatient cardiovascular disease, headache, upper respiratory disease, GI disease, musculoskeletal problems, eye disease, sleep disorders, and miscellaneous topics. This chapter continues that outpatient orientation but emphasizes longitudinal prevention and older-adult care. The NBME content outline also highlights health maintenance, diagnosis, management, and older adults as recurring themes, making this material a practical endpoint for the internal medicine book.

1. Ambulatory longitudinal care framework

Ambulatory medicine is not simply inpatient medicine performed more slowly. The clinician must manage uncertainty over time, prevent disease before it appears, monitor chronic illness, reconcile medications, identify subtle decline, and create follow-up plans that are safe even when a diagnosis is incomplete. The core outpatient questions are: What is the patient worried about today? Is there an emergency hiding behind a common complaint? What preventive care is due? What chronic diseases are uncontrolled? What medications are helping, harming, or redundant? What functional or social barriers make the plan unrealistic? What interval and trigger should prompt reassessment?

For a study book and future Graph-RAG system, ambulatory care can be represented as repeated cycles: assess risk, screen only when benefit exceeds harm, treat modifiable risk factors, monitor response, deprescribe when harms exceed benefits, and update goals of care as disease burden and function change. The same condition may have different management in a healthy 45-year-old, a frail 88-year-old with falls, or a patient with advanced dementia and limited prognosis.

Ambulatory task	What the clinician is deciding	Common data elements	Common trap
Triage	Can this be managed outpatient or does it need ED/inpatient care?	Vital signs, appearance, oxygenation, mental status, pain severity, organ-threat symptoms, support at home	Treating a red flag as routine clinic disease because the complaint is common
Prevention	Which screening, vaccines, and risk-reduction steps are due?	Age, sex anatomy, pregnancy potential, smoking, ASCVD risk, family history, prior screening, immune status	Doing tests because they are available rather than because they change outcomes
Chronic care	Is the disease controlled and is the treatment tolerated?	Home readings, labs, adherence, adverse effects, complications, patient goals	Escalating medication without checking technique, adherence, cost, diet, or adverse effects
Medication review	Which drugs should be continued, stopped, simplified, or monitored?	Indication, dose, renal function, interactions, falls, cognition, anticholinergic burden	Continuing medicines indefinitely after the original indication has disappeared
Geriatric assessment	What function, cognition, mobility, and support shape the plan?	ADLs, IADLs, gait, falls, cognition, mood, hearing/vision, nutrition, caregiver status	Using disease-specific targets without considering frailty or life expectancy
Palliative care	How should care align with values, prognosis, and symptom burden?	Goals, capacity, surrogate decision-maker, code status, symptom scale, hospice eligibility	Confusing palliative care with giving up or delaying symptom relief until the last days of life

2. Prevention and health maintenance

Preventive medicine is a structured attempt to prevent premature morbidity and mortality while avoiding overdiagnosis, false positives, unnecessary procedures, and treatment burden. Screening is most useful when a disease has an asymptomatic phase, the test is accurate enough, early treatment improves outcomes, and the patient has enough life expectancy and functional reserve to benefit. Screening is less useful when the patient would not want or tolerate downstream treatment, when a positive result would not change management, or when harms of false positives outweigh likely benefit.

Preventive care should be personalized but not random. A practical annual review includes tobacco and nicotine use, alcohol and substance use, diet, physical activity, BMI/waist pattern, blood pressure, lipid/ASCVD risk, diabetes risk, depression, intimate partner violence when appropriate, fall risk in older adults, sexual health, contraception or pregnancy planning when relevant, dental/vision/hearing needs, vaccines, and age-appropriate cancer screening.

Screening / prevention domain	Typical adult approach	Numbers and thresholds to remember	Clinical caveats
Blood pressure	Measure accurately and confirm elevated readings with repeated office or home/ambulatory measurements when feasible	Normal <120/<80; elevated 120-129/<80; stage 1 130-139 or 80-89; stage 2 ≥140 or ≥90 by ACC/AHA framework	Do not diagnose from a single casual reading unless severe hypertension or end-organ damage is present; check technique and white-coat/masked HTN
Diabetes	Screen adults with overweight/obesity plus risk factors and broadly by age/risk per current guidance	A1c 5.7-6.4 prediabetes; ≥6.5 diabetes if confirmed; fasting glucose 100-125 prediabetes, ≥126 diabetes if confirmed	Earlier screening for prior gestational diabetes, PCOS, strong family history, high-risk ethnicity, or vascular disease
Lipids / ASCVD	Estimate 10-year ASCVD risk for primary prevention and identify diabetes, LDL ≥190, clinical ASCVD, and risk enhancers	LDL ≥190 mg/dL generally high-intensity statin; adults 40-75 use ASCVD risk and diabetes status	Risk discussion includes absolute benefit, adverse effects, cost, pregnancy potential, and patient preference
Colorectal cancer	Screen average-risk adults starting at 45; continue routinely through 75; individualize 76-85	Colonoscopy every 10 yr, annual FIT, stool DNA-FIT every 1-3 yr, CT colonography every 5 yr are common options	Positive stool-based test requires colonoscopy; do not order stool test if colonoscopy would not be pursued
Breast cancer	Mammography based on age/risk and shared decision-making	Common U.S. frameworks screen biennially from 40 to 74; high-risk genetic/family history may need earlier MRI-based strategy	Consider life expectancy, prior results, breast density, family history, and patient values
Cervical cancer	Cytology and/or HPV-based screening by age	21-29: cytology every 3 yr; 30-65: cytology q3 yr, high-risk HPV q5 yr, or cotest q5 yr	Stop after 65 if adequately screened and not high-risk; do not screen after total hysterectomy for benign disease with cervix removed
Lung cancer	Annual low-dose CT for high-risk smoking history	Age 50-80, ≥20 pack-years, current smoker or quit within 15 years in USPSTF framework	Only screen if patient is healthy enough and willing to undergo curative treatment; combine with smoking cessation
Osteoporosis	DXA screening in older women and selected younger/high-risk adults	Women ≥65; postmenopausal women <65 if fracture risk elevated; evaluate men with risk factors	Fragility hip or vertebral fracture can establish treatment need regardless of DXA value
Depression	Use validated tools when systems exist for diagnosis and treatment	PHQ-2 then PHQ-9 common; assess suicidality when positive	Symptoms overlap with grief, sleep disorders, hypothyroidism, medications, dementia, and substance use
Tobacco	Ask every visit, advise quitting, offer pharmacotherapy and counseling	5 A framework: Ask, Advise, Assess, Assist, Arrange	Varenicline, NRT, and bupropion are options; match to contraindications and patient preference

3. Adult vaccination and immunization safety

Vaccination is one of the most effective ambulatory interventions. The practical approach is to review age-based vaccines, risk-based vaccines, immune status, pregnancy status, asplenia, chronic lung/heart/liver/kidney disease, diabetes, occupational exposure, travel, and immunosuppressive medications. Live vaccines require special caution in pregnancy and significant immunosuppression; inactivated and recombinant vaccines are generally safer but may be less immunogenic during intense immunosuppression.

Vaccine area	High-yield adult pattern	Special considerations
Influenza	Annual vaccination for adults; especially important in older adults, pregnancy, chronic disease, and immunocompromise	Egg allergy rarely prevents vaccination; timing is usually fall/winter but local circulation matters
COVID-19	Follow current CDC schedule for age, risk, prior vaccination, and immune status	Guidance changes; check current schedule for high-risk and immunocompromised patients
Tdap/Td	One adult Tdap if not previously received, then Td or Tdap booster every 10 yr; Tdap each pregnancy	Dirty wound management may require booster depending on timing and vaccine history
Zoster recombinant vaccine	Recombinant zoster vaccine series for adults ≥50 and for immunocompromised adults ≥19	Not live; give even with prior shingles, but avoid during acute zoster episode
Pneumococcal	Age- and risk-based PCV strategy for adults, with specific choices depending on prior vaccine history	Chronic heart/lung/liver disease, diabetes, smoking, CSF leak, cochlear implant, asplenia, and immunocompromise change indications
RSV	Older adults and selected high-risk adults based on current ACIP/CDC recommendations	Shared decision and risk-based thresholds have evolved; check current schedule

Vaccine area	High-yield adult pattern	Special considerations
Hepatitis B	Universal adult vaccination through age 59; age ≥ 60 if risk factor or request depending guidance	Diabetes, CKD/dialysis, liver disease, sexual exposure risk, injection drug use, incarceration, and health-care exposure increase priority
HPV	Routine through age 26 if not adequately vaccinated; shared decision age 27-45	Prevents new HPV infection; does not treat existing infection
MMR/Varicella	Assess immunity in adults without evidence of immunity	Live vaccines: avoid in pregnancy and severe immunosuppression
Travel/asplenia/special risk	Meningococcal, Hib, hepatitis A, typhoid, yellow fever, Japanese encephalitis, rabies depending risk	Coordinate with travel clinic; immunosuppression may alter live vaccine options

4. Geriatric assessment: function before diagnosis

Geriatrics begins with the recognition that chronological age is less important than physiologic reserve, function, cognition, medication burden, social support, and goals. Two adults who are 82 years old can have completely different care plans: one may benefit from aggressive risk-factor modification and cancer screening; another with advanced dementia, recurrent falls, and limited life expectancy may benefit more from deprescribing, fall prevention, caregiver support, and symptom-focused care. Functional assessment is therefore not extra; it is part of the diagnosis.

Activities of daily living (ADLs) are basic self-care tasks such as bathing, dressing, toileting, transferring, continence, and feeding. Instrumental activities of daily living (IADLs) are higher-level tasks such as managing medications, finances, transportation, shopping, cooking, housekeeping, and using communication devices. Loss of IADLs often precedes loss of ADLs and may reveal early cognitive decline, depression, sensory impairment, neurologic disease, medication toxicity, or unsafe living conditions.

Geriatric domain	What to ask or examine	Why it matters	Action when abnormal
ADLs	Bathing, dressing, toileting, transferring, continence, feeding	Predicts need for caregiver support, rehab, home health, facility care	Assess safety, caregiver strain, equipment, PT/OT, home services
IADLs	Medications, finances, transportation, shopping, cooking, phone/technology	Early marker of cognitive or functional decline	Medication simplification, cognitive screen, social work, caregiver involvement
Mobility	Gait speed, Timed Up and Go, chair rise, balance, assistive device use	Falls, hospitalization, mortality, independence	PT, strength/balance training, home safety, vision/footwear review
Cognition	Mini-Cog, MoCA/MMSE when indicated, collateral history	Medication adherence, driving, finances, capacity, delirium risk	Workup reversible causes; diagnose dementia subtype; plan support and safety
Mood	PHQ-2/PHQ-9, anxiety screen, grief, loneliness	Depression mimics dementia and worsens chronic disease control	Treat depression, assess suicide risk, therapy/social connection
Sensory	Hearing, vision, dental status	Falls, isolation, delirium, medication errors, nutrition	Audiology, glasses/cataracts, dental care, communication adaptation
Nutrition	Weight trend, appetite, chewing/swallowing, food access	Frailty, sarcopenia, pressure injury, mortality	Treat reversible causes; protein/calorie support; evaluate dysphagia or malignancy when indicated
Social support	Caregiver, housing, transportation, finances, advance directive	Determines whether outpatient plans are feasible	Social work, home health, caregiver resources, safety planning

5. Frailty, falls, syncope overlap, and mobility decline

Frailty is a state of reduced physiologic reserve in which minor stressors cause disproportionate decline. It is not the same as disability, comorbidity, or age, although they overlap. Frailty should be suspected with unintentional weight loss, exhaustion, weakness, slow gait, low activity, recurrent falls, poor recovery from illness, and repeated hospitalizations. Frailty changes management because preventive tests and tight disease targets may provide less benefit and more harm when life expectancy and reserve are limited.

Falls are sentinel events. They may reflect environmental hazards, orthostatic hypotension, sedatives, neuropathy, vision impairment, vestibular disease, Parkinsonism, stroke, arrhythmia, aortic stenosis, dehydration, infection, delirium, foot pain, alcohol use, or muscle weakness. The outpatient evaluation should not stop at “mechanical fall” unless the history clearly supports a trip/slip and there is no syncope, presyncope, focal neurologic deficit, chest pain, palpitations, seizure activity, new weakness, or injury requiring imaging.

Fall assessment element	High-yield details	Intervention
History of event	Before/during/after symptoms, witness, loss of consciousness, head strike, anticoagulants, seizure features	ED/imaging threshold if head trauma, anticoagulation, neuro deficit, prolonged LOC, severe injury
Orthostatic vitals	Supine/seated and standing BP/HR; symptoms with standing	Hydration, medication adjustment, compression, slow position changes; evaluate autonomic failure if persistent

Fall assessment element	High-yield details	Intervention
Medication review	Benzodiazepines, Z-drugs, anticholinergics, opioids, antipsychotics, antihypertensives, hypoglycemics, alcohol	Deprescribe or reduce dose; simplify regimen; treat pain/sleep/anxiety with safer strategies
Gait/balance	Timed Up and Go, gait speed, chair rise, assistive device fit	PT, strength/balance exercise, cane/walker optimization
Vision/feet/home	Cataracts, bifocals, neuropathy, footwear, rugs, stairs, lighting, bathroom hazards	Vision correction, podiatry, footwear, home safety modifications, grab bars
Bone health	Fragility fracture history, vitamin D/calcium intake, DXA risk	Osteoporosis treatment if indicated; fall prevention plus fracture prevention
Cardiac/neurologic clues	Syncope, palpitations, exertional symptoms, focal deficits, Parkinsonism	ECG, rhythm monitoring, echo, neuro evaluation guided by features

6. Delirium, dementia, and depression

Delirium is an acute disturbance in attention and awareness with fluctuating course. It is a medical emergency in older adults until proven otherwise. Dementia is chronic cognitive decline that interferes with function. Depression can mimic dementia, worsen cognition, and coexist with neurodegenerative disease. The clinical task is to establish time course: hours to days with fluctuation suggests delirium; months to years with progressive IADL/ADL impairment suggests dementia; prominent low mood, anhedonia, guilt, sleep/appetite change, and subjective concentration difficulty suggests depression, although overlap is common.

Delirium triggers include infection, hypoxia, dehydration, pain, urinary retention, constipation, sleep deprivation, new medications, withdrawal, metabolic derangements, renal/hepatic failure, stroke, seizure, myocardial infarction, and environmental change. The first treatment is identification and correction of causes plus nonpharmacologic management: reorientation, sleep/wake protection, mobilization, sensory aids, hydration/nutrition, pain control, bowel/bladder care, and family presence. Antipsychotics do not treat delirium itself and should be reserved for severe agitation threatening safety after reversible causes and nonpharmacologic measures are addressed.

Feature	Delirium	Dementia	Depression
Onset	Acute: hours to days	Insidious: months to years	Variable; often weeks to months
Course	Fluctuating, worse at night, inattention prominent	Progressive decline; attention often preserved until later	May fluctuate with mood; effort and motivation affected
Consciousness/attention	Altered level of arousal, impaired attention	Usually normal arousal early	Usually normal arousal; concentration impaired
Hallucinations	Common, especially visual in delirium or Lewy body disease	May occur in Lewy body/Parkinson dementia or late disease	May occur if psychotic depression but not typical
Workup priority	Search for acute medical cause: infection, hypoxia, drugs, metabolic, CNS, pain/retention/constipation	Cognitive testing, functional history, medication review, labs for reversible contributors, neuroimaging when indicated	PHQ-9, suicide risk, grief/substance/medical contributors
Treatment emphasis	Treat cause, supportive environment, avoid restraints/sedatives when possible	Safety, caregiver support, risk-factor control, selected cognitive meds, advance planning	Psychotherapy, antidepressants when indicated, sleep/activity/social connection

Do-not-miss delirium principle

A new change in attention, behavior, mobility, appetite, continence, or function in an older adult can be the presenting symptom of infection, hypoxia, medication toxicity, metabolic disease, stroke, myocardial infarction, urinary retention, constipation, or pain. Do not attribute acute confusion to baseline dementia without collateral history.

7. Polypharmacy, deprescribing, and medication safety

Polypharmacy is not simply the number of medications; it is the use of medications whose cumulative harms, interactions, monitoring burden, or complexity exceed benefit. Older adults are vulnerable because renal clearance, hepatic metabolism, body composition, homeostatic reserve, cognition, vision, dexterity, and medication management capacity change. A medication can be appropriate in one older adult and inappropriate in another depending on indication, dose, prognosis, adverse effects, and goals.

A deprescribing visit asks: What is each medication for? Is the indication still present? Is there evidence of benefit for this patient's time horizon? Is the dose appropriate for kidney/liver function? Is it causing falls, confusion, constipation, urinary retention, bleeding, hypoglycemia, orthostasis, anorexia, sedation, or interactions? Can the regimen be simplified? Which drug would the patient most like to stop? Some drugs can be stopped abruptly, while others require tapering, such as benzodiazepines, beta-blockers, clonidine, opioids, gabapentinoids, antidepressants, corticosteroids, and antiseizure medications.

High-risk medication group in older adults	Common harms	Safer strategy / caution
Benzodiazepines and Z-drugs	Falls, fractures, delirium, cognitive impairment, motor vehicle crashes, dependence	Avoid for chronic insomnia/anxiety when possible; taper gradually; CBT-I and safer anxiety treatments
Strong anticholinergics	Confusion, dry mouth, constipation, urinary retention, blurred vision, falls	Reduce anticholinergic burden; consider non-drug or less anticholinergic alternatives
Opioids	Sedation, constipation, falls, respiratory depression, dependence, delirium	Use lowest effective dose, bowel regimen, naloxone when appropriate, functional goals, avoid benzodiazepine combination
NSAIDs	GI bleeding, kidney injury, hypertension, heart failure exacerbation	Use topical NSAIDs/acetaminophen/PT when possible; gastroprotection/risk review if necessary
Sulfonylureas/insulin overly tight control	Hypoglycemia, falls, seizures, hospitalization	Relax A1c targets in frailty/limited life expectancy; prefer lower hypoglycemia risk drugs
Antipsychotics	Stroke/mortality risk in dementia, parkinsonism, sedation, QT prolongation	Reserve for psychosis or dangerous agitation when non-drug measures fail; reassess frequently
Anticoagulants/antiplatelets	Bleeding, drug interactions, fall-related hemorrhage	Do not stop solely for fall risk; reassess indication, renal dose, bleeding risk, goals
Proton pump inhibitors chronic use	C. difficile risk, hypomagnesemia, fractures association, B12/iron issues in selected patients	Continue for strong indication; step down if uncomplicated GERD controlled

8. Urinary incontinence, pressure injuries, nutrition, sarcopenia, and osteoporosis

Geriatric syndromes often arise from multiple small insults rather than one disease. Urinary incontinence can result from detrusor overactivity, pelvic floor weakness, prostate obstruction, medications, diabetes, neurologic disease, constipation, delirium, UTI, mobility impairment, and environmental barriers. Pressure injuries result from pressure, shear, immobility, malnutrition, moisture, poor perfusion, and sensory impairment. Sarcopenia and malnutrition worsen falls, immune function, wound healing, medication toxicity, and mortality. Osteoporosis converts falls into fractures; fracture prevention therefore requires both bone therapy and fall prevention.

Problem	Pattern recognition	Initial management
Urge incontinence	Sudden urge, frequency/nocturia, leakage before reaching toilet	Bladder training, timed voiding, reduce bladder irritants, treat constipation, pelvic floor therapy; cautious meds if needed
Stress incontinence	Leak with cough/laugh/exertion	Pelvic floor exercises, weight loss if appropriate, pessary/surgical referral for selected patients
Overflow incontinence	Dribbling, weak stream, incomplete emptying, high postvoid residual	Evaluate obstruction, medications, neuropathy; avoid anticholinergics until retention excluded
Functional incontinence	Cannot reach toilet due to mobility/cognition/environment	Scheduled toileting, bedside commode, mobility aids, caregiver support, environmental modification
Pressure injury risk	Immobility, malnutrition, incontinence/moisture, neuropathy, poor perfusion	Repositioning, pressure redistribution surfaces, moisture control, nutrition/protein support, wound care
Sarcopenia/frailty nutrition	Weight loss, low strength, low intake, food insecurity, dysphagia, dental disease	Protein/calorie optimization, resistance exercise, treat depression/dysphagia/dental issues, social food support
Osteoporosis	Fragility fracture, low BMD, glucocorticoids, low body weight, smoking, falls	Calcium/vitamin D adequacy, weight-bearing/resistance exercise, fall prevention, bisphosphonate or other therapy if indicated

9. Palliative care, hospice, capacity, and goals of care

Palliative care is specialized care focused on symptom relief, communication, decision support, and quality of life for patients with serious illness. It can be provided alongside disease-directed therapy and is not limited to the final days of life. Hospice is a specific model of care generally used when life expectancy is limited, usually around six months if the disease follows its expected course, and when the focus has shifted primarily to comfort rather than curative treatment. Many patients benefit from palliative principles long before they qualify for hospice.

Goals-of-care conversations translate medical facts into patient-centered plans. A useful structure is: assess understanding, share prognosis in plain language, ask what matters most, identify unacceptable health states, discuss tradeoffs, recommend a plan, and document the decision-maker and code status. Capacity is decision-specific: the patient must communicate a choice, understand relevant information, appreciate how it applies to them, and reason about options. Lack of capacity for one decision does not imply global incapacity.

Concept	Definition / key point	Common clinical application
Palliative care	Symptom-focused and goal-concordant care for serious illness at any stage	Advanced heart failure, COPD, cancer, dementia, cirrhosis, CKD, recurrent hospitalizations
Hospice	Comfort-focused care for limited prognosis, commonly ≤ 6 months if illness follows expected course	Stop nonbeneficial monitoring, emphasize home support, symptom kits, caregiver education
Code status	Preference for CPR/intubation in cardiopulmonary arrest or respiratory failure context	Should be connected to prognosis and goals; avoid presenting CPR as a neutral technical choice
Advance directive	Document naming surrogate and/or stating preferences	Review before crises; ensure surrogate understands patient values
Capacity	Ability to communicate, understand, appreciate, and reason about a specific decision	Assess when refusing high-benefit care, accepting high-risk care, delirium, dementia, psychosis, intoxication
Informed consent	Benefits, risks, alternatives, and likely consequences of no treatment	Requires capacity, voluntariness, and adequate information
Surrogate decision-making	Uses substituted judgment first, best interest if patient preferences unknown	Clarify legal hierarchy by jurisdiction and document discussions

10. Symptom control in palliative medicine

Palliative symptom management should be proactive, measurable, and paired with explanation. Pain, dyspnea, nausea, constipation, anxiety, delirium, secretions, insomnia, anorexia, and fatigue are common final common pathways of many diseases. The goal is not to sedate by default; it is to relieve suffering while preserving the patient's preferred level of alertness when possible. Opioids are appropriate for moderate to severe pain and refractory dyspnea in serious illness when titrated carefully and monitored for constipation, sedation, nausea, and respiratory risk.

Symptom	Initial evaluation	Treatment framework
Pain	Type: nociceptive, neuropathic, bone, visceral; severity, function, renal/hepatic status, prior opioid exposure	Acetaminophen/topicals for mild pain; opioids for moderate-severe serious illness pain; adjuvants for neuropathic/bone pain; bowel regimen with opioids
Dyspnea	Hypoxemia, bronchospasm, effusion, edema, PE, anxiety, anemia, terminal disease	Treat reversible cause if consistent with goals; fan/positioning; oxygen if hypoxemic; low-dose opioid for refractory dyspnea
Nausea	Constipation, obstruction, meds, uremia, hypercalcemia, vestibular, intracranial, gastroparesis	Match antiemetic to mechanism: dopamine antagonists, serotonin antagonists, antihistamines/anticholinergics, prokinetics, steroids in selected cases
Constipation	Opioids, immobility, dehydration, obstruction, hypercalcemia, anticholinergics	Stimulant laxative plus osmotic agent often needed with opioids; avoid bulk fiber in poor intake/obstruction risk
Delirium/agitation	Pain, urinary retention, constipation, infection, hypoxia, meds, metabolic disease	Non-drug measures and cause correction; antipsychotic only for severe distress or safety threat when consistent with goals
Anorexia/cachexia	Disease stage, dysphagia, depression, nausea, constipation, medication effects	Treat reversible causes; counsel family that forced feeding may not improve comfort in advanced disease; consider appetite meds selectively
Terminal secretions	Noisy pooled secretions near end of life; often more distressing to family than patient	Reposition, reduce fluids if appropriate, anticholinergic agents selectively, family education

11. Integrated outpatient algorithms

Preventive visit algorithm

1) Clarify agenda and acute concerns. 2) Review medications, allergies, problem list, outside records, and home measurements. 3) Update vital signs and functional/social context. 4) Address tobacco, alcohol, diet, exercise, sleep, sexual health, safety, mood, and violence risk when appropriate. 5) Check vaccines. 6) Check cancer and chronic disease screening due dates. 7) Reconcile ASCVD/diabetes/osteoporosis/fall risk. 8) Choose only interventions with likely benefit for this patient. 9) Document follow-up interval and return precautions.

Scenario	Key first questions	High-yield workup	Management emphasis
Older adult with fall	Syncope? Head strike? Anticoagulant? New weakness? Infection symptoms? Medication change? Prior falls?	Orthostatics, gait/balance, vision/feet, med review; ECG/labs/imaging if clues	Prevent injury, treat cause, deprescribe fall-risk meds, PT, home safety, bone health
Older adult with confusion	Acute or chronic? Fluctuating? Baseline cognition? New meds? Pain/retention/constipation? Infection/hypoxia?	Vitals/O ₂ , glucose, medication review, UA only if urinary/systemic symptoms, CBC/CMP as indicated, neuro exam	Treat delirium cause; avoid sedatives/restraints; involve family and sensory aids

Scenario	Key first questions	High-yield workup	Management emphasis
Patient requesting cancer screening with limited prognosis	What outcome would they want? Would they undergo biopsy/surgery/chemo? Prior screening?	Estimate functional status and life expectancy; review harms/benefits	Avoid low-value screening when downstream treatment would not be pursued; focus on symptoms and comfort
Polypharmacy visit	What is each med for? What can be stopped? Falls/confusion/constipation? Cost/adherence?	Renal/hepatic dosing, drug interactions, anticholinergic/sedative burden	Deprescribe one step at a time; taper high-risk meds; monitor withdrawal/recurrence
Advanced illness visit	What do they understand? What matters most? What tradeoffs are acceptable? Who is surrogate?	Symptom assessment, prognosis communication, capacity assessment	Recommend goal-concordant plan, document code status and advance care planning, consider palliative/hospice referral

12. High-yield red flags and treatment traps

Area	Red flag	Why it matters
Prevention	Screening test ordered when patient would not accept or benefit from downstream treatment	Creates harm through false positives, procedures, anxiety, and cascades without benefit
Vaccines	Live vaccine given during pregnancy or severe immunosuppression	Can cause serious vaccine-related infection or fetal risk depending vaccine
Falls	Fall labeled mechanical without asking about syncope, medications, orthostasis, head strike, or neurologic symptoms	Misses arrhythmia, stroke, intracranial bleed, medication toxicity, or seizure
Delirium	Acute confusion attributed to dementia or aging	Misses infection, hypoxia, drug toxicity, metabolic disease, retention, constipation, MI, stroke
Dementia	No collateral history or functional assessment	Cognitive score alone cannot establish real-world impairment or safety needs
Depression	No suicide assessment after positive screen	Misses immediate safety threat
Polypharmacy	Continuing benzodiazepines/Z-drugs/anticholinergics in recurrent falls or confusion	Potentially reversible cause of injury and cognitive impairment
Diabetes in frailty	Tight glycemic control despite falls, hypoglycemia, dementia, or limited life expectancy	Hypoglycemia harm may exceed microvascular prevention benefit
Palliative care	Waiting until active dying to discuss goals or symptoms	Delays comfort, planning, and caregiver support
Capacity	Assuming disagreement means incapacity	Capacity requires structured assessment; patients with capacity may choose risky options

References and source framework

- Uploaded source framework: ambulatory Step-Up chapter packet and Step-Up outline, which place ambulatory medicine and appendix skills near the end of the internal medicine structure.
- National Board of Medical Examiners. Internal Medicine Subject Exam Content Outline: health maintenance, diagnosis, management, and older adults are emphasized domains.
- Centers for Disease Control and Prevention. Adult Immunization Schedule by Age and Addendum, 2025-2026 online schedule pages.
- U.S. Preventive Services Task Force. A and B Recommendations; colorectal, cervical, lung, depression, tobacco, and related adult preventive service recommendation pages.
- American Geriatrics Society. 2023 Updated AGS Beers Criteria for Potentially Inappropriate Medication Use in Older Adults.
- Centers for Disease Control and Prevention. STEADI algorithm and clinical resources for fall risk screening, assessment, and intervention in community-dwelling older adults.
- Core palliative care principles synthesized from standard internal medicine, geriatrics, and hospice/palliative care educational frameworks.